## HOW WE TESTED

We decided to segregate all drives on the basis of pricing, effectively doing away with the traditional divisions based on technology or capacity. After all, users want the best solution for any given price. Higher performance drives will generally tend to be costlier than power saving drives, so this slight inverse delta (performance/price) works well. We did distinguish between SSDs and HDDs to avoid comparing the proverbial apples to oranges. Typically we use a single test rig for any test to ensure uniformity of results. However, this time we had a couple of unique contestants, both from Seagate. The first, their Barracuda XS drive is based on a SATA 3.0 interface, a new standard promising theoretical transfers up to 600 MBps. Naturally we had to use a SATA 3.0 board, and there are precious few of those. We also got Seagate's 600 GB Cheetah based on an SAS (Serial Attached SCSI) connectivity. Few motherboards support SAS connects, so we ended up with three different rigs.

The Barracuda XS was tested under both SATA 2.0 and 3.0 conditions, and the results (spoiler!) surprised us. We credit ASUS and Western Digital in particular — the former for supplying us their typically excellent boards for use in this test and the latter for providing us with Velociraptor drives to use as references. Intel also deserves mention for their provision of three of the best processors on their top three platforms.

We continue to use Velociraptors over SSD, because of the high write speeds these drives give in RAID. Clean installs were done on each of the platforms and two partitions created on the RAID 0 array, the primary one of a size of 100 GB, while the second partition consisted of the remaining space. All extra Windows functions that potentially slow down access, such as System Restore, File Indexing and a few unused services were disabled. Post installation of benchmarks, each partition was cleaned, defragmented and then we run a disk scan.

For theoretical benchmarks, we used HD Tach 3.0.1.0 and SiSoft Sandra 2010 with the latest update pack. Both benchmarks require unformatted drives to perform write testing. After these benchmarks were complete we formatted the test drives and created two partitions of equal size.

For our real world tests we used an 8 GB .RAR file to simulate sequential operation to the disk and a folder of size 8 GB consisting of a whole lot of different types of files of varying sizes; this to simulate an random or non-sequential operation. These files were copied to and fro from the second partition on the RAID 0 array to the test drives. A copy to the test drive simulates a write operation, while a copy back to the RAID 0 array simulates a read.

We also did these tests intra drive, that is copying from one partition of the test drive to the other. This is a read + write operation, which is more taxing and slower as you will see. Finally we used Adobe PhotoShop CS 2 to test the amount of time taken to load a 1 GB image. Both the image and the scratch disk were kept on the test drives, on the same partition, effectively simulating file access. In case of hard drives, we gave no weight to Features, since there were none that need considering before buying an HDD. For SSDs, we gave points for SSDs that featured Trim, a command that basically allows the SSD controller to zero in on clean blocks and add them to the pool of free blocks. These blocks represent deleted files, making Trim somewhat of a "trash compacter" command.

**TEST RIGS**

| | Main Rig | SATA 3.0 Rig | SAS Rig |
|---|---|---|---|
| Motherboard | ASUS P5Q3 Deluxe | ASUS P7P55D-E Premium | ASUS P6T Deluxe |
| Processor | Intel Core 2 Extreme QX9650 (3 GHz) | Intel Core i7 870 (2.93 GHz) | Intel Core i7 975 Extreme (3.33 GHz) |
| RAM | 2 x 2 GB DDR3 1600 MHz (1333 MHz @ 8-8-8-20) | 2 x 2 GB DDR3 1600 MHz (1333 MHz @ 8-8-8-20) | 3 x 1 GB DDR 2000 MHz (1333 MHz @ 8-8-8-20) |
| HDD | 2 x Western Digital Velociraptor (RAID 0) | 2 x Western Digital Velociraptor (RAID 0) | 2 x Western Digital Velociraptor (RAID 0) |
| Graphics Card | Palit GeForce GTS 250 | Palit GeForce GTS 250 | Palit GeForce GTS 250 |
| OS | Windows Vista Premium 32-bit | Windows Vista Premium 32-bit | Windows Vista Premium 32-bit |

our assumptions on present HDD spindle technology we reckon manufacturers can squeeze out another 10 per cent of performance in real life scenarios, given the same spindle speeds, meaning there's very little headroom to exploit a theoretical limit of 300 MBps, let along 600 MBps (SATA 3.0).


But there was one other solution that crushed the WD 2 TB Black Edition and the mighty Velociraptor WD3000HLFS — Seagate's Cheetah 15K.7 ST3600057SS, was a shocker, and one of the meanest things this side of a platter that your cash can buy. To be honest, it's a little unfair to compare what is, essentially an enterprise-grade device to these drives, but Seagate has been smart to offer this drive on an SAS-interface, rather than the de facto SCSI interface that such 15,000 rpm drives regularly feature. SAS drives are rare, as are SAS controllers (especially on desktop boards), but we dug one up, the ASUS P6T Deluxe. The Cheetah lives up to its namesake when it comes to raw throughput, particularly the write tests, where the faster spindle speed is definitely speeding up things. The read scores are no less impressive and this drive a sequential read score of 180 megabytes per second — very impressive, and by far, the fastest hard drive we've come across. While people will claim that this figure is child's play for SSDs, the write speed of 142.2 MBps proves its mettle — if you want the fastest drive out there for a desktop system, this is the one to go shopping for, although you'll